Full-Range Feature Extraction Network Based on Quality–Quantity-Balance Sample Enhancement for Hyperspectral Image Classification

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***Abstract*— Hyperspectral remote sensing images exhibit fine spectral curves, but they are also susceptible to spectral variations caused by factors such as cloud and haze. It is evident that these issues become more pronounced when there is a limited number of labeled samples available. Thus, a full range feature extraction network (FRFENet) based on quality–quantity-balance sample enhancement is proposed for hyperspectral image classification (HSIC). First, the full-range feature extraction method combines local-range, short-range, and long-range spatial–spectral features to address spectral variability and ensure accurate feature extraction, particularly in scenarios with limited labeled samples. Furthermore, the approach of balancing quality and quantity for pseudolabeled samples allows for an increased number of pseudolabels while maintaining their quality, effectively leveraging unlabeled samples. In addition, the utilization of superpixel region homogeneity directly contributes to an expanded training sample set, resulting in improved classification performance of the algorithm. Experiments on three hyperspectral image (HSI) encompass datasets indicate that the FRFENet can obtain better classification performance when compared with the other ten state-of-the-art methods.**

***Index Terms*— Convolutional neural network (CNN), full range feature extraction, graph convolution, hyperspectral image classification (HSIC), sample enhancement.**

I. Introduction

HYPERSPECTRAL images (HSIs) encompass numerous spectral bands, offering a wealth of distinctive features surpassing those of RGB images [1], [2]. As a result, HSIs find extensive applications in various domains, such as precision agriculture, military surveillance, and wetland research [3]. Furthermore, hyperspectral image classification (HSIC) plays a vital role by providing detailed category information at the pixel level, facilitating subsequent tasks [4], [5], [6], [7].

Regarding feature extraction capabilities, algorithms for HSIC can be categorized into two main types: machine learning (ML)-based and deep learning-based approaches [8], [9], [10]. Specifically, ML algorithms leverage statistical learning and probabilistic methods to acquire the inherent statistical patterns within a hyperspectral image, enabling the construction of classification models. However, ML-based algorithms have demonstrated two primary limitations when applied to high-dimensional hyperspectral data. First, they heavily rely on hand-crafted features or expert knowledge, and second, they often lack the necessary capacity to extract discriminative, abstract, and robust high-level features. These features represent advanced concepts or patterns within the data, which is crucial in aiding ML models to better understand the information within the image and subsequently perform classification.

Deep learning-based methods have emerged as the dominant approach in HSIC due to their remarkable capability for feature representation [11], [12]. Within deep learning, various techniques have been developed, including deep belief networks (DBNs) [13], convolutional neural networks (CNNs) [4], deformable convolutional networks (DCNs) [10], [14], vision transformer-based networks (ViTs) [15], [16], [17], and graph convolutional networks (GCNs) [18]. These algorithm networks offer advanced modeling and learning capabilities, allowing for more effective and accurate analysis of hyperspectral data in the field of HSIC [19], [20].

Among them, CNN and GCN have gained significant attention in the research community due to their exceptional feature extraction capabilities [21], [22], [23], [24], [25]. Specifically, CNN excels at extracting local features and capturing texture information by utilizing convolution and activation operations. It effectively learns the spatial and spectral relationships within local regions, enabling it to represent edge, texture, and local structural features accurately.

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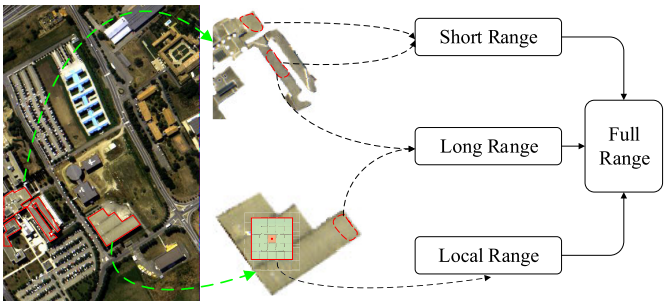


Fig. 1. Illustration among local range, short range, long range, and full range.

For example, different-dimensional convolutional networks, such as 1DCNN, 2DCNN, and 3DCNN, along with residual-based networks, have emerged in the field of HSIC [26], [27]. Subsequently, there have been developments in DCNs and hybrid networks that combine convolutional operations of different dimensions. However, the limited receptive field of CNN restricts its capability to capture long-range contextual information. Consequently, it may encounter challenges in handling global consistency and spatial continuity features. To address this limitation, a combination of CNN and GCN becomes essential in order to leverage both local and global features [28], [29], [30]. In addition, in order to reduce the computational complexity, it is common to preprocess hyperspectral images using clustering or superpixel segmentation algorithms, and this preprocessing step can significantly reduce the local perception capability of GCN [31]. Therefore, GCN is unable to accomplish pixel-level local information extraction [18], [32]. In summary, CNN primarily captures spatial–spectral features within a local range, while GCN focus on capturing spatial–spectral features within either short or long ranges [33]. As a result, both CNN and GCN have inherent limitations in terms of feature extraction. To better illustrate the relationships between local-range features, short-range features, and long-range features, Fig. 1 is displayed as follows.

In general, the local range can be understood as the receptive field of the CNN's convolutional kernel, as shown in Fig. 1. And, the short range refers to the convolution range of 1–3 hops in GCN; the long range refers to the convolution range of distant hops in GCN. At the same time, the extraction of long-range features in GCN refers to the ability to perceive global node information in a single aggregation operation [34], [35], [36]. Furthermore, there are some limitations of feature extraction at a single range. Specifically, the isolated utilization of local-range feature extraction may impose constraints on the treatment of global consistency and spatial continuity features. Solely relying on short-range feature extraction may result in an inadequate consideration of local intricacies and small-scale characteristics. Similarly, employing long-range feature extraction in isolation may neglect the comprehensive integration of local spatial–spectral information.

Therefore, there have been research efforts aimed at integrating these various feature extraction approaches. Specifically, Liu et al. [28] proposed a CNN-enhanced GCN network that integrates the capabilities of both CNN and GCN, demonstrating excellent classification performance. Subsequently, Zhu et al. [37] introduced a long–short graph convolution approach that combines long-range and short-range graph convolutions, enabling the extraction of both long-distance and short-distance information. In addition, Liu et al. [33] achieved pixel-level and node-level joint convolutions at the convolutional layer, enabling the perception of local-range features and short-range features. From the perspective of hyperspectral data, during the data collection process, various factors, such as illumination and atmospheric conditions, can introduce spectral variations. These spectral variations significantly interfere with the extraction of features for different land cover categories, leading to deviations in the extracted features and impacting the classification performance of algorithms. In addition, in practical applications, it can be challenging to obtain a sufficient number of labeled samples to train algorithm models. The scarcity of labeled samples undoubtedly exacerbates the impact of spectral variations on classification algorithms.

Therefore, inspired by the aforementioned methods, it is worth exploring how to effectively integrate local-range, short-range, and long-range features under the condition of insufficient labeled samples, in order to mitigate the impact of spectral variations. Furthermore, semi-supervised methods are also commonly used to address the issue of insufficient labeled samples [38], [39], [40], [41]. Based on confidence-based pseudolabeling, the semi-supervised classification algorithm for hyperspectral image leverages a common approach in semi-supervised learning [42], [43], [44]. This method utilizes a small amount of labeled data to train a network and then generates pseudolabels for unlabeled data based on the network's predictions and their corresponding confidence scores. By leveraging pseudolabels, this approach incorporates unlabeled data into the training process, expanding the size of the training set and improving the performance of the classifier. However, the accuracy of the pseudolabels is crucial for the algorithm's effectiveness. Therefore, it is essential to design appropriate strategies for generating and filtering pseudolabels to enhance their reliability and reduce errors [45].

Nowadays, there have been many research works on how to set the threshold for pseudolabeling [46], [47], [48]. The correctness of pseudolabels has a significant impact on model training. Good prediction results contribute to the convergence of the model and the ability to learn new patterns, while poor prediction results can interfere with the model's training process [41], [49]. In general, the selection of the threshold directly affects two indicators of pseudolabeling: quality and quantity [50]. The balance between quality and quantity is one of the research focuses. Specifically, a high confidence threshold ensures the quality of pseudolabels but significantly reduces the quantity of pseudolabels that meet the threshold. For example, Sohn [51] generated pseudolabels for unlabeled images based on model predictions with weak data augmentation and only selected samples with high confidence. They enforced consistent predictions on strongly augmented images using consistency regularization.

In addition, while a low confidence threshold can introduce a larger number of unlabeled samples, the low confidence directly reduces the quality of these samples. This may

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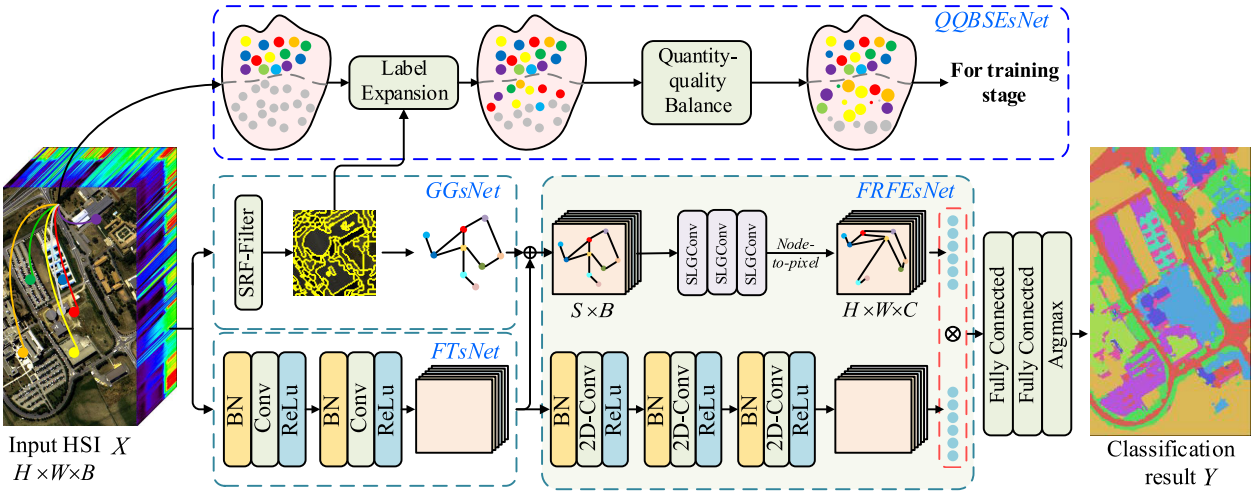


Fig. 2. Flowchart of the FRFENet framework. ⊕ means combining the encoded graph nodes with their corresponding features. ⊗ denotes the feature concatenation. "Node-to-pixel" indicates that the node-level features are converted into pixel-level for classification. SRF-Filter will be introduced in the proposed method. S represents the number of superpixels after SRF-Filter.

introduce samples with incorrect labels, further disrupting the training of the network and decreasing classification performance. Therefore, a low confidence threshold is not directly used. Instead, dynamic thresholds or class-specific thresholds are often adopted to address this issue. For example, Xu et al. [52] innovatively proposed a method for semi-supervised learning by using dynamic thresholding to select unlabeled samples. They improved the selection strategy for unlabeled samples during the training process by using dynamically changing thresholds to choose more effective unlabeled samples for training. Zhang et al. [48] proposed curriculum pseudolabeling (CPL), which involves adjusting the thresholds for different classes flexibly at each time step, allowing informative unlabeled data and their pseudolabels to pass through. Subsequently, in order to balance the relationship between the quality and quantity of pseudolabels, Chen et al. [50] proposed a novel method called SoftMatch. By maintaining high-quantity and high-quality pseudolabels during the training process, SoftMatch effectively leverages unlabeled data to overcome the trade-off issue. Zhao et al. [53] proposed a multiattention transformer and adaptive superpixel segmentation-based active learning method (MAT-ASSAL) to obtain pseudolabels by using superpixel segmentation and active learning strategies and obtained advanced classification performance. However, its performance depends on the setting of the superpixel segmentation scale and the number of active learning iterations.

Based on the analysis above, to enhance the accuracy of high-dimensional image classification in the case of limited labeled samples, a full range feature extraction network (FRFENet) based on quality–quantity-balance sample enhancement is proposed for HSIC. Specifically, in order to ensure a sufficient quantity of pseudolabels and improve the quality of pseudolabels, quality–quantity-balanced sample enhancement subnetwork (QQBSEsNet) is proposed to enhance the sample. In this way, pseudolabels with sufficient quantity and high quality can be selected at one time. In addition, to address the spectral variability in high-dimensional image data, full-range feature extraction subnetwork (FRFEsNet) is proposed to make full use of the full range spatial–spectral features of HSIs, which not only consider the spectral information of pixels in the image but also consider their spatial distribution. This advantage can more comprehensively describe the characteristics of the ground objects in the image, rather than just the spectral information of a certain pixel, so as to better distinguish between different classes.

The main contributions of this article are as follows.

1) First, to solve the problem of lack of labeled samples in HSIC and find a general sample expansion strategy, a classification framework based on quality–quantity-balanced sample enhancement is proposed. This framework can take into account the quality and quantity of pseudolabels and extract high-quality features effectively, thus improving the performance of classification algorithms.
2) Second, FRFEsNet combines local-range, short-range, and long-range spatial–spectral features to achieve full-range feature extraction, which can address spectral variability and ensure accurate feature extraction in the case of insufficient labeled samples.
3) Third, QQBSEsNet can expand the quantity of pseudolabels while ensuring their quality, effectively utilizing unlabeled samples. In addition, the homogeneity of superpixel regions is also utilized to directly increase the number of training samples, further enhancing the classification performance of the algorithm.

The following structure of this article is as follows. Section II introduces the methodology of the proposed method. Section III presents the classification results and parameter discussion. Section IV summarizes this article.

II. Proposed Method

The flowchart of the FRFENet is shown in Fig. 2, which contains four subnetworks, graph generation subnetwork (GGsNet), feature transformation subnetwork (FTsNet),

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FRFEsNet, and QQBSEsNet. First, the HSI is input into GGsNet for superpixel segmentation, and the obtained superpixel regions are encoded into the graph. Note that the SRF-Filter in GGsNet can obtain more homogeneous multiscale superpixel regions, which will be briefly introduced. Then, FTsNet is used to reduce the dimensionality of the raw HSI, resulting in deep and robust spectral features. Subsequently, the graph and the features processed by FTsNet are combined to obtain node-level features. Finally, the pixel-level and node-level features are input into FRFEsNet, which utilizes both short–long graph convolution (SLGConv) and 2-D convolution to extract spatial–spectral features at different ranges and obtains the final classification result. Note that the QQBSEsNet increases the training sample quantity by extending pseudolabels, which plays a role throughout the entire training process of the network.

The input HSI and the output result of FRFENet are represented as $\mathbf{X} = \{\boldsymbol{x}_i\}_{i=1}^{I} \in \mathbb{R}^{H\times W\times B}$ and $\mathbf{Y} = \{y_i\}_{i=1}^{I} \in \mathbb{R}^{H\times W}$, respectively, where H, W, B, and I represent the height, width, dimensionality, and total number of pixels in the input HSI, respectively. In this notation, $\boldsymbol{x}_i$ represents a pixel in the HSI, and y_i represents the corresponding pixel in the classification result map. The details of each subnetwork within FRFENet will be explained in Section II.

A. QQBSEsNet

In practical applications, the acquisition of labeled samples often incurs significant manpower and financial costs, leading to insufficient availability of labeled samples. It is important to note that the adequacy of labeled samples is not only solely determined by their quantity but also rather by whether the classification accuracy of the algorithm meets the requirements of the specific application at a given number of labeled samples. Therefore, in order to improve the classification accuracy of the algorithm when labeled samples are limited, the QQBSEsNet includes two approaches for sample augmentation: superpixel region label expansion (SRLE) and quality–quantity balance expansion (QQBE).

1) Superpixel Region Label Expansion: SRLE primarily expands the quantity of labeled samples by leveraging the high homogeneity of superpixel regions based on the assumption that all pixels within the same superpixel region belong to the same class. By assigning corresponding labels to the entire superpixel region based on the labels of the selected training samples and randomly selecting a certain number of pseudolabels within that superpixel region, it is possible to effectively increase the quantity of labeled samples and provide more training data for the algorithm. The diagram of SRLE is shown in Fig. 3.

In order to ensure the high homogeneity of superpixel regions, the SRF-Filter is introduced to process the hyperspectral image segmented by the simple linear iterative clustering (SLIC) algorithm. The detail of the SRF-Filter can be seen in [8]. The segmented HSI X is composed of many superpixel regions R_m. The superpixel region R_m contains many pixels, which is represented by the elements Ω and x_n

$$R_m \rightarrow : (\Omega, x_n) \tag{1}$$

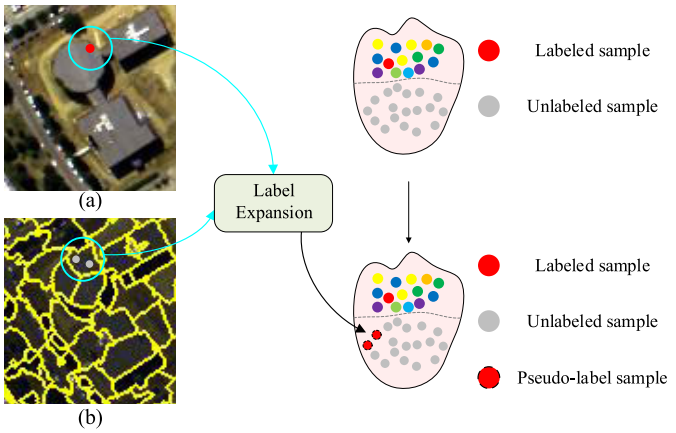


Fig. 3. Diagram of the SRLE. (a) HSI and (b) superpixel region. Different colors represent samples of different classes, gray represents the unlabeled samples, and colored samples represent the labeled samples.

where Ω denotes the boundary of region R_m and x_n denotes the nth pixel in the superpixel region R_m.

The training samples selected from labeled samples can be represented as x_n^{tr}, and the unlabeled samples can be represented as x_n^{un}. Then, the label expansion process of SRLE can be represented as follows:

$$R_m\left(x_n^{\text{tr}}, x_n^{un}\right) \rightarrow R_m\left(x_n^{\text{tr}}, k * x_n^{\text{pse}}\right) \tag{2}$$

where x_n^{pse} represents the pseudolabel selected based on x_n^{tr}, k denotes the number of planned pseudolabels to be selected within the superpixel region R_m.

2) Quality–Quantity Balance Expansion: When using pseudolabels, the confidence threshold can be considered as a step function where pseudolabels with confidence above the threshold are considered correct, while others are considered incorrect. In order to ensure the quality of pseudolabels, a high confidence threshold is typically set, which reduces the number of eligible pseudolabels. Conversely, to increase the quantity of pseudolabels, the quality of pseudolabels may decrease, introducing erroneous pseudolabels. Research has shown that the fundamental reason behind the quantity–quality trade-off is the lack of sophisticated assumptions imposed by the weighting function on the distribution of pseudolabels [50]. In addition, in the field of HSIC, effectively utilizing pseudolabels is also a challenge. Therefore, we introduce this idea, which is described in detail below.

Based on the previous analysis, QQBE is proposed to balance the relationship between the quality and quantity of pseudolabels. The core idea is to derive a truncated Gaussian function from the assumption on the marginal distribution to fit the confidence distribution. It assigns lower weights to pseudolabels that deviate from the mean of the Gaussian in terms of their confidence, indicating that they are more likely to be correct. The parameters of the Gaussian function are estimated using the historical predictions from the model during the training process. This approach achieves the goal of maintaining high quantity and high quality of pseudolabels during training. The diagram of QQBE is shown in Fig. 4.

Assume that D_L as $\{x_l^i, y_l^i\}_{i=1}^{n_l}$ represents the pixel x_l with the label $y_s \in \{1, 2, \ldots, C\}$ (C is the category number), and

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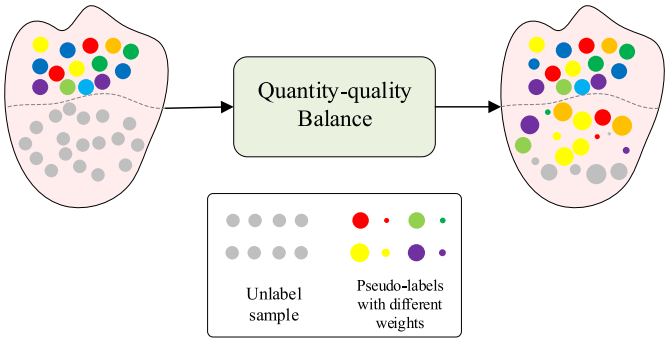


Fig. 4. Diagram of the QQBE. Circles with different sizes represent different weights. The larger the circle, the greater the weight, and the smaller the circle, the smaller the weight. And, different colors represent different categories.

D_U as $\{x_u^j\}_{j=1}^{n_u}$ represents the pixel x_u without the label, where n_l and n_u represent the number of the pixel of the labeled and unlabeled training sample.

During the training process, the loss function $\mathcal{L}$ is composed of supervised loss $\mathcal{L}_l$ and unsupervised loss $\mathcal{L}_u$, which can be represented as follows:

$$\mathcal{L} = \mathcal{L}_l + \mathcal{L}_u. \tag{3}$$

The loss function used in the FRFENet algorithm is the cross-entropy loss, and it incorporates category number weights [54]. The expression of the loss function is as follows:

$$\mathcal{L}_l = \frac{1}{B_l}\sum_{i=1}^{B_l}\sum_{c=1}^{C} N_c^{-1}\mathbf{O}_c\left(y_i^l\right)\log\left(\mathbf{Z}_c^p\left(x_i^l\right)\right) \tag{4}$$

where N_c represents the number of samples in category c and B_l represents the number of training samples in each batch. The term $\mathbf{O}_c(y_i)$ denotes the one-hot encoding format of y_i, and $\mathbf{Z}_c^p$ represents the predicted probability of sample $\boldsymbol{x}_i$ belonging to the cth category

$$\mathcal{L}_u = \frac{1}{B_u}\sum_{i=1}^{B_u}\sum_{c=1}^{C}\lambda(\mathbf{p}_i)\mathbf{O}_c\left(y_i^u\right)\log\left(\mathbf{Z}_c^p\left(x_n^{\text{pse}}\right)\right) \tag{5}$$

where B_u represents the batch size for unlabeled data and x_n^{pse} denotes the pseudolabel. $\lambda(\mathbf{p})$ is the sample weight function, ranging from 0 to $\lambda_{\max}$.

From a theoretical perspective, an optimal sample weight function $\lambda(\mathbf{p})$ should accurately reflect the true distribution and achieve a balance between the quantity and quality of pseudolabels. In order to better fit the data distribution, we assume that the underlying probability mass function (PMF) $\lambda(\mathbf{p})$ of the marginal distribution follows a dynamic truncated Gaussian distribution with mean μ_t and variance σ_t at the tth training iteration. Then, $\lambda(\mathbf{p})$ can be expressed as follows:

$$\lambda(\mathbf{p}) = \begin{cases} \lambda_{\max}\exp\left(-\dfrac{(\max(\mathbf{p})-\mu_t)^2}{2\sigma_t^2}\right), & \text{if } \max(\mathbf{p}) < \mu_t \\ \lambda_{\max}, & \text{otherwise .} \end{cases} \tag{6}$$

Subsequently, to generate the mean and variance in (6) adaptively, the mean and variance of the next iteration can be inferred based on the model's historical data. Specifically, it can be expressed as follows:

$$\hat{\mu}_b = \hat{\mathbb{E}}_{B_u}\left[\max(\mathbf{p})\right] = \frac{1}{B_u}\sum_{i=1}^{B_u}\max(\mathbf{p}_i)$$
$$\hat{\sigma}_b^2 = \hat{\text{Var}}_{B_u}\left[\max(\mathbf{p})\right] = \frac{1}{B_u}\sum_{i=1}^{B_u}\left(\max(\mathbf{p}_i)-\hat{\mu}_b\right)^2. \tag{7}$$

Then, the momentum m from previous batches is aggregated to obtain a more stable estimate

$$\hat{\mu}_t = m\hat{\mu}_{t-1} + (1-m)\hat{\mu}_b$$
$$\hat{\sigma}_t^2 = m\hat{\sigma}_{t-1}^2 + (1-m)\frac{B_u}{B_u-1}\hat{\sigma}_b^2 \tag{8}$$

where $\hat{\mu}_0$ and $\hat{\sigma}_0^2$ can be initialized as $(1/C)$ and 1, respectively.

The spectrum of pixels of the same class in HSIs is different, so the fact that pixels are highly homogeneous within the same superpixel does not mean that they are the same. At the same time, the QQBE strategy selected samples of high quality effectively. Therefore, QQBSEsNet can obtain both quantity and quality of pseudolabel samples, rather than simple extended duplicate samples.

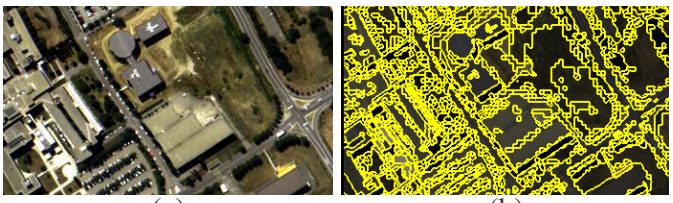


Fig. 5. (a) HSI. (b) Part of the segmentation result maps.

B. GGsNet and FTsNet

The ground objects in hyperspectral images exhibit significant variations in size, and the segmentation results from a single-scale superpixel method cannot fully capture the rich spatial structural information. Therefore, in GGsNet, the SRF-Filter is utilized to fuse superpixel regions [8]. Fig. 5 illustrates the fused superpixel regions, where it can be observed that the size of superpixel regions varies according to the actual size of ground objects. This fusion approach ensures both accurate ground object boundaries and high homogeneity within the regions. Then, each superpixel region is treated as a node, and a graph structure is constructed based on the adjacency relationship between superpixel regions.

The FTsNet is introduced to reduce the spectral dimension, extract deep spectral features, and denoise the spectral information. The process of FTsNet convolution can be expressed as follows:

$$\mathbf{X}' = \sigma\left(\boldsymbol{W}_{\text{FTs}}^2 \cdot \text{BN}\left(\sigma\left(\boldsymbol{W}_{\text{FTs}}^1 \cdot BN(\mathbf{X})\right)\right)\right) \tag{9}$$

where $\text{BN}(\bullet)$ and $\sigma(\bullet)$ denote the feature normalization and the activation function, $\boldsymbol{W}_{\text{FTs}}^1$ and $\boldsymbol{W}_{\text{FTs}}^2$ denote the weight of the first and second convolution kernels, respectively. And, the

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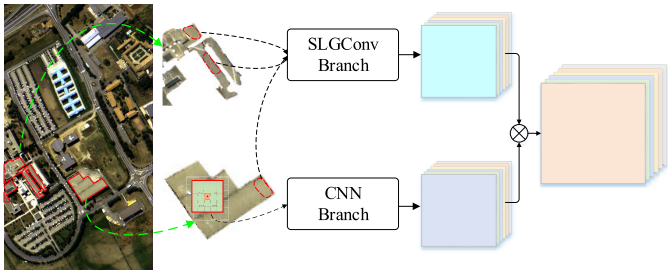


Fig. 6. Diagram of the FRFEsNet.

size of the output feature $\mathbf{X}' = \{x'_1, x'_2, \ldots, x'_{HW}\}$ is $p \times q \times B'$.

Subsequently, the graph structure outputted by GGsNet is combined with the pixel-level features $\mathbf{X}^{\text{out}}$ from FTsNet to obtain node-level features. Specifically, the node feature is obtained by taking the average of the pixel features within each superpixel region.

C. FRFEsNet

Compared with optical RGB images, hyperspectral images have the advantage of fine spectral resolution. However, due to their high spectral resolution, the spectral curves of the same object can be easily affected by unfavorable imaging conditions, such as clouds and haze, resulting in significant variations in the spectral curves of the same object and possible similarities between spectral curves of different objects.

In the case of limited labeled samples, it is imaginable that certain objects may have only a few or a dozen labeled samples. A small number of labeled samples cannot capture the full spectrum variation. Consequently, the model fails to learn all the variations of the objects, leading to incorrect classification. Therefore, an FRFEsNet is proposed, as shown in Fig. 6.

It can be observed that FRFEsNet mainly consists of two branches: the SLGConv branch and the CNN branch. The SLGConv branch is responsible for extracting short-range and long-range features, while the CNN branch is responsible for extracting local features. Subsequently, the features extracted from both branches are concatenated together as the final feature output.

1) SLGConv Branch: The SLGConv branch mainly consists of SLGConv, which combines the features extracted from long-range graph convolution and short-range graph convolution through weighted fusion. SLGConv retains the advantages of both types of graph convolutions and compensates for their respective shortcomings, enabling accurate feature extraction in both long range and short range [37].

If we represent the node-level features as $\mathbf{H}$, the SLGConv can be defined as follows:

$$\mathbf{H}^{\text{out}} = \text{FC}\left[\left(\mathbf{C}_{\text{long}}\, \mathbf{H}\mathbf{W}^{12}_{fc}\right) \otimes \left(\mathbf{C}_{\text{short}}\, \mathbf{H}\mathbf{W}^{12}_{fc}\right)\right] \tag{10}$$

where FC[•] denotes the fully connected layer in SLGConv, $\otimes$ denotes feature concatenation, and $\mathbf{W}^{12}_{fc}$ are the parameters. Formula (10) achieves the feature extraction operation of SLGConv. Note that $\mathbf{C}_{\text{long}}$ and $\mathbf{C}_{\text{short}}$ are the weights for the long-range graph convolution and short-range graph convolution, respectively.

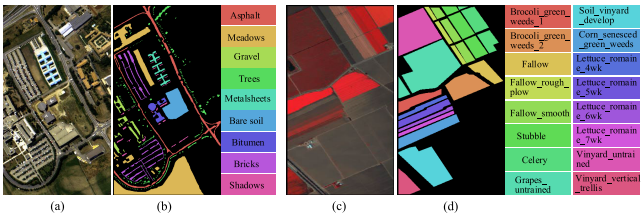


Fig. 7. Hyperspectral images and ground truth of the PaviaU and Salinas datasets. (a) and (c) False-color image. (b) and (d) Ground truth.

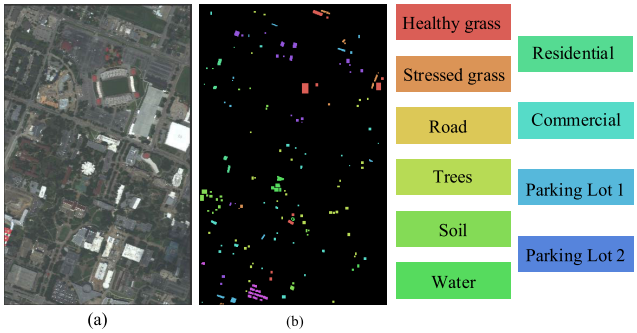


Fig. 8. Hyperspectral images and ground truth of the Houston2013 dataset. (a) False-color image. (b) Ground truth.

2) CNN Branch: The CNN branch mainly includes normalization, 2-D convolution, pooling, activation functions, and other operations. With the advantages of local receptive fields and weight sharing, CNN can accurately extract features within a local distance range.

The pixel-level features are denoted by $\mathbf{X}'$, the CNN branch can be defined as follows:

$$\mathbf{X}^{\text{out}} = \sigma\left(\boldsymbol{W}^3 \cdot \sigma\left(\boldsymbol{W}^2 \cdot \left(\sigma\left(\boldsymbol{W}^1 \cdot \mathbf{X}'\right)\right)\right)\right). \tag{11}$$

Subsequently, the final output of FRFEsNet can be represented as follows:

$$\mathbf{Z}^{\text{out}} = \left[\mathbf{X}^{\text{out}} \otimes \mathbf{H}^{\text{out}}\right]. \tag{12}$$

III. Experiments

In this section, datasets, associated experimental settings, experimental results, and analysis are presented.

A. HSI Datasets

The performance of FRFENet is evaluated on three datasets: Pavia University (PaviaU) dataset [54], Salinas dataset [55], and Houston University 2013 (Houston2013) dataset [56]. These datasets are chosen to assess the classification accuracy of FRFENet. The hyperspectral images and ground truth of the three datasets are shown in Figs. 7 and 8.

The PaviaU dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS). The ROSIS sensor is capable of acquiring 115 spectral bands within a wavelength range, with a spectral resolution of. The PaviaU dataset has a spatial resolution of 1.3 m and consists of 103 spectral bands. The size of the dataset is 610 × 340 pixels, and it contains nine labeled land cover classes.

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The Salinas dataset was captured by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) in the Salinas Valley region of California. The AVIRIS sensor captures 204 spectral bands within a wavelength range. The Salinas dataset has a spatial resolution of 3.7 m and a size of 512 × 217 pixels. It consists of 16 labeled land cover classes.

The Houston2013 dataset was captured using the CASI-1500 sensor from ITRES, Canada. It contains 144 spectral bands within a wavelength range and has a spatial resolution of 2.5 m. The Houston2013 dataset has a size of 344 × 552 pixels and exhibits complex characteristics, including a large number of ground objects, overlapping distributions, and varying sizes. In particular, the Houston2013 dataset is a part (from rows 3 to 346 and columns 635 to 1186) of the original Houston dataset, which contains most of the ground objects of the original dataset.

B. Experimental Setup

Ten state-of-the-art algorithms are selected for comparison, including support vector machine (SVM) [57], K-nearest neighbors (KNN) [58], 2DCNN, double-branch dual-attention mechanism (DBDA) [59], HybridSN [60], ViT [61], SpectralFormer [62], MAT-ASSAL [53], CNN-enhanced graph convolutional network (CEGCN) [28], MiniGCN [63], and multiscale dynamic graph convolutional network (MDGCN) [34]. To quantitatively evaluate the performance of the algorithm, the following evaluation indicators commonly used in HSI tasks are employed: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) [64]. These indicators provide a quantitative assessment of the algorithm's classification performance.

FRFENet is implemented using the PyTorch framework in Python, utilizing the Adam optimizer with an initial learning rate of 0.0005. The algorithm is trained for a total of 600 epochs. The hardware setup for the implementation includes a Xeon Silver 4210R CPU, a GTX-3090 GPU, and 49 GB of RAM memory. The input patch size for the CNN-based algorithm is set to 17 × 17. The parameter k in (2) is set to 2.

To evaluate the classification accuracy with minimal training samples, a random selection of 0.5%, 0.1%, and 99.4% of samples per class from the PaviaU and Salinas datasets are used as training, validation, and testing datasets, respectively. Due to the small size of some classes in the Houston2013 dataset, 1%, 1%, and 98% of samples per class from the Houston2013 dataset are used as training, validation, and testing datasets, respectively. Detailed category information of the three datasets is shown in Tables I–III. It is important to note that the number of training samples is standardized across all algorithms to ensure a fair comparison. The experimental results presented in this section are based on ten parallel repetitions of the experiments, providing robustness and reliable statistical analysis.

C. Analysis of Classification Results

In this section, we evaluate and compare the classification performance of all algorithms. The classification result maps for the PaviaU, Salinas, and Houston2013 datasets are illustrated in Figs. 9–11, respectively. Furthermore, detailed tables (Tables IV–VI) present the classification accuracy for each category, providing a comprehensive analysis of the algorithms' performance on these datasets.

For the PaviaU dataset, the proposed algorithm FRFENet achieves the highest overall classification accuracy of 98.72% with only 0.5% of training samples. Table IV demonstrates that the classification accuracy of ML-based algorithms is generally low and insufficient to handle high-dimensional hyperspectral data with limited training samples. Among the CNN-based algorithms, only the DBDA algorithm with an attention mechanism achieves a classification accuracy of 97.65%. Transformer-based algorithms require a large number of training samples to optimize parameters; hence, they do not achieve high classification performance. In contrast, GCN-based algorithms, especially CEGCN and

TABLE I
CATEGORY INFORMATION OF THE PAVIAU DATASET

Class ID	Class Name	Train.	Val.	Test.
1	Asphalt	34	7	6590
2	Meadows	94	19	18536
3	Gravel	11	3	2085
4	Trees/Grass	16	4	3044
5	Metalsheets	7	2	1336
6	Baresoil	26	6	4997
7	Bitumen	7	2	1321
8	Bricks	19	4	3659
9	Shadows	5	1	941
	Total	219	48	42509

TABLE II
CATEGORY INFORMATION OF THE SALINAS DATASET

Class ID	Class Name	Train.	Val.	Test.
1	Brocoli_green_weeds_1	11	3	1995
2	Brocoli_green_weeds_2	19	4	3703
3	Fallow	10	2	1964
4	Fallow_rough_plow	7	2	1385
5	Fallow_smooth	14	3	2661
6	Stubble	20	4	3935
7	Celery	18	4	3557
8	Grapes_untrained	57	12	11202
9	Soil_vinyard_develop	32	7	6164
10	Corn_senesced_green_weeds	17	4	3257
11	Lettuce_romaine_4wk	6	2	1060
12	Lettuce_romaine_5wk	10	2	1915
13	Lettuce_romaine_6wk	5	1	910
14	Lettuce_romaine_7wk	6	2	1062
15	Vinyard_untrained	37	8	7223
16	Vinyard_vertical_trellis	10	2	1795
	Total	279	62	53788

TABLE III
CATEGORY INFORMATION OF THE HOUSTON2013 DATASET

Class ID	Class Name	Train.	Val.	Test.
1	Healthy grass	7	7	657
2	Stressed grass	3	3	233
3	Road	4	4	343
4	Trees	7	7	621
5	Soil	5	5	482
6	Water	4	4	331
7	Residential	3	3	229
8	Commercial	5	5	469
9	Parking Lot 1	6	6	544
10	Parking Lot 2	5	5	391
	Total	49	49	4300

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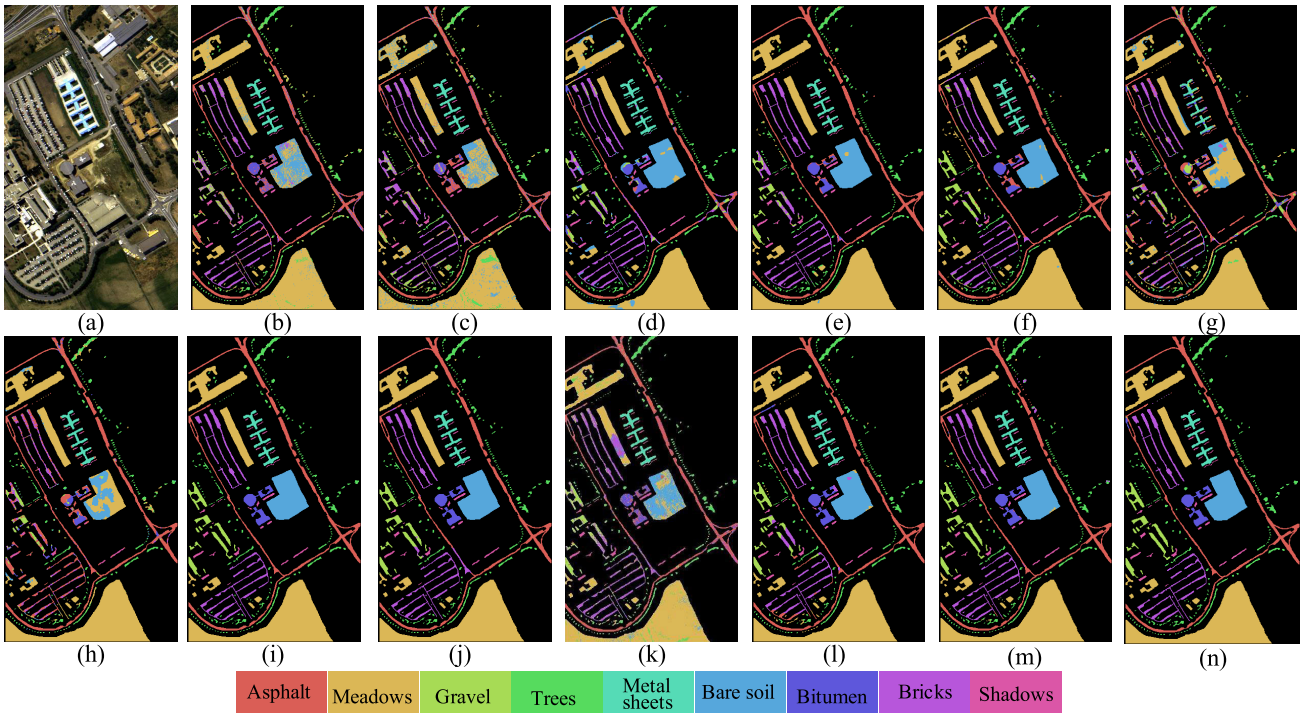


Fig. 9. Classification maps of all different methods on the PaviaU dataset. (a) False-color image. (b) SVM. (c) KNN. (d) 2DCNN. (e) DBDA. (f) HybridSN. (g) ViT. (h) SpectralFormer. (i) MAT-ASSAL. (j) CEGCN. (k) miniGCN. (l) MDGCN. (m) FRFENet. (n) Ground truth.

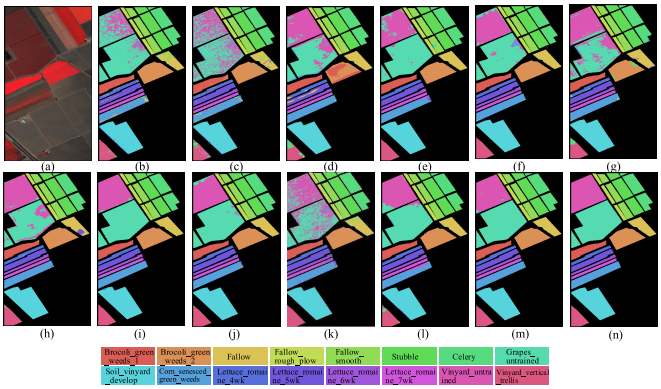


Fig. 10. Classification maps of all different methods on the Salinas dataset. (a) False-color image. (b) SVM. (c) KNN. (d) 2DCNN. (e) DBDA. (f) HybridSN. (g) ViT. (h) SpectralFormer. (i) MAT-ASSAL. (j) CEGCN. (k) miniGCN. (l) MDGCN. (m) FRFENet. (n) Ground truth.

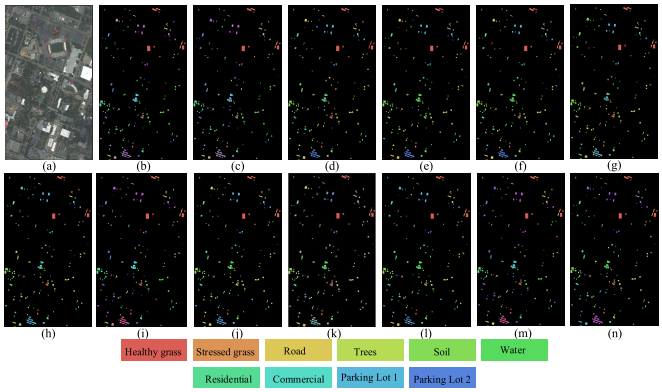


Fig. 11. Classification maps of all different methods on the Houston2013 dataset. (a) False-color image. (b) SVM. (c) KNN. (d) 2DCNN. (e) DBDA. (f) HybridSN. (g) ViT. (h) SpectralFormer. (i) MAT-ASSAL. (j) CEGCN. (k) miniGCN. (l) MDGCN. (m) FRFENet. (n) Ground truth.

MDGCN, achieve good classification performance with their semi-supervised training approach. As for the FRFENet algorithm, the sample enhancement method and FRFENet significantly enhance its classification performance, enabling it to effectively address the spectral variability in hyperspectral data.

Fig. 9 presents the classification result maps of various algorithms. It can be observed from Fig. 9(m) that there are hardly any region-level misclassifications or pixel-level misclassifications in the classification result maps. This also reflects that the proposed algorithm corrects misclassified pixels caused by spectral variability through the extraction of long-range, short-range, and local-range features. In addition, the label sample enhancement strategy employed by FRFENet increases the number of available training samples and ensures the proper optimization of network parameters (NPs). On the other hand, some algorithms exhibit varying degrees of region-level misclassifications due to spectral variability, particularly the KNN algorithm and ViT algorithm.

For the Salinas dataset, FRFENet achieved the highest classification accuracy of 99.14%. Fig. 10(m) displays the classification result map of the FRFENet algorithm. The classification accuracy and result map demonstrate that FRFENet outperforms other algorithms in terms of classification performance. To address the misclassification issue of the "Grapes untrained" and "Vinyard untrained" categories, the proposed algorithm utilizes a full-range feature extraction strategy to correct this problem. Among the comparative algorithms, CEGCN achieved the second-highest classification

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TABLE IV

CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE PAVIAU DATASET

Class	ML-based		CNN-based			Transformer-based			GNN-based			
	SVM	KNN	2DCNN	DBDA	HybridSN	ViT	SpectralFormer	MAT-ASSAL	CEGCN	MiniGCN	MDGCN	FRFENet
1	85.04±5.25	76.73±0.53	94.36±2.65	95.89±2.50	95.67±2.21	70.65±8.77	85.47±6.70	96.95±2.01	**98.84±1.26**	79.95±7.65	98.10±1.08	98.53±1.81
2	93.86±3.07	92.06±4.88	98.23±1.39	99.35±0.38	99.82±0.15	89.66±7.60	93.06±5.58	98.95±0.99	**99.91±0.22**	90.86±3.11	99.99±0	99.23±0.46
3	52.51±15.22	28.44±16.45	59.57±6.45	96.26±3.61	88.03±5.95	38.10±7.18	38.09±9.10	98.01±2.22	88.31±10.33	70.99±7.45	77.03±5.80	**99.63±0.02**
4	75.09±6.50	49.68±16.63	91.61±3.78	98.44±1.02	82.70±3.25	73.99±2.79	79.45±6.37	**94.56±4.21**	93.84±2.15	89.77±0.83	90.63±0.88	93.35±2.50
5	99.24±0.31	98.34±0.62	97.82±2.44	99.13±1.02	99.90±0.19	75.63±8.30	82.32±6.89	97.92±2.03	**100±0**	99.33±0.28	98.85±0.42	99.85±0.17
6	65.59±11.38	30.88±15.06	89.89±13.59	99.35±0.60	98.19±1.13	51.95±17.19	54.87±14.08	97.56±4.21	**99.44±1.69**	69.99±7.55	97.47±0.67	99.24±0.04
7	40.01±18.39	65.02±7.35	53.95±18.73	99.29±1.37	93.03±7.75	44.69±16.26	57.67±12.73	93.99±5.37	99.72±0.51	67.82±2.35	97.79±3.84	**99.97±0.01**
8	79.10±4.66	83.13±5.60	89.68±8.45	89.82±4.21	80.59±5.30	67.50±0.40	56.16±9.44	97.79±1.46	98.41±1.83	65.6±17.17	92.10±2.70	**99.15±0.04**
9	99.68±0.13	99.41±0.59	84.46±9.00	98.81±1.26	80.22±16.24	21.00±16.66	50.37±5.95	95.36±1.49	98.27±1.99	**99.89±0.09**	97.24±0	97.60±5.15
OA	83.15±1.30	75.08±1.19	91.84±2.54	97.65±0.56	94.88±0.75	73.36±2.26	78.16±1.99	97.76±1.67	98.52±0.79	83.24±2.61	96.76±0.57	**98.72±0.41**
AA	76.68±3.59	69.30±4.18	84.40±4.03	97.37±0.45	90.90±2.24	59.24±4.29	66.38±1.24	96.78±2.05	97.42±1.41	81.58±1.50	94.35±1.11	**98.51±0.68**
Kappa	77.31±1.89	65.81±2.31	89.09±3.55	96.88±0.75	93.16±1.01	64.17±2.35	70.37±2.35	97.01±2.23	98.03±1.05	77.91±3.16	95.69±0.75	**98.31±0.55**

TABLE V

CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE SALINAS DATASET

Class	ML-based		CNN-based			Transformer-based			GNN-based			
	SVM	KNN	2DCNN	DBDA	HybridSN	ViT	SpectralFormer	MAT-ASSAL	CEGCN	MiniGCN	MDGCN	FRFENet
1	96.79±1.30	93.97±0.98	82.81±24.62	99.71±0.55	**100±0**	90.95±7.69	97.13±3.09	**100±0**	99.91±0.26	98.90±0.19	99.50±0	**100±0**
2	97.65±1.69	98.01±0.28	84.50±19.85	99.97±0.05	**100±0**	93.19±0.76	98.20±1.93	**100±0**	**100±0**	99.29±0.24	99.92±0.19	99.85±0.21
3	87.19±7.97	85.46±9.28	87.78±8.99	96.32±0.24	99.82±0.17	85.48±9.51	89.81±7.53	95.91±0.94	**100±0**	59.49±18.87	100±0	98.64±2.64
4	95.43±3.56	98.73±0.36	97.13±4.15	90.89±2.65	89.42±7.14	95.32±2.23	94.91±3.36	97.53±5.54	99.28±0.69	98.72±0.74	**99.65±0.13**	98.13±2.81
5	95.83±2.41	96.66±0.82	94.93±4.22	99.33±0.30	**99.77±0.41**	98.12±1.30	98.99±1.43	98.69±1.34	99.19±0.64	96.47±2.82	97.88±0.02	98.18±0.42
6	98.80±0.37	98.96±0.25	99.43±0.98	99.03±0.69	99.70±0.45	99.78±0.06	98.25±1.46	**100±0**	99.95±0.14	99.78±0.02	99.55±0.12	99.91±0.11
7	99.25±0.32	98.73±0.43	96.42±6.82	99.00±1.60	99.90±0.06	97.32±2.71	98.96±0.71	**100±0**	99.99±0.02	99.48±0.07	95.61±3.74	99.68±0.27
8	88.48±1.18	74.41±5.24	87.47±18.15	96.53±3.49	93.09±1.22	68.54±5.38	82.10±1.02	78.75±0.87	97.40±2.53	70.59±5.08	95.39±2.11	**99.30±0.94**
9	96.95±0.83	96.91±1.52	97.22±2.68	99.39±0.19	99.95±0.09	98.12±2.51	99.97±0.04	**100±0**	**100±0**	99.64±0.24	99.74±0.35	99.97±0.03
10	83.39±4.13	76.64±5.51	95.64±2.13	96.28±2.15	98.99±0.68	95.80±1.91	95.52±0.42	94.32±0.98	96.21±0.88	92.13±1.22	89.51±2.52	**98.41±1.36**
11	79.65±4.42	72.93±17.44	87.83±13.92	89.80±3.71	93.04±3.15	86.92±8.40	94.51±3.33	95.31±1.09	**99.68±0.48**	94.66±1.49	95.76±1.13	98.93±1.29
12	99.04±1.40	98.33±1.17	82.31±24.81	99.45±0.66	**99.98±0.04**	97.20±2.55	99.10±1.28	91.60±2.57	99.98±0.05	99.67±0.36	100±0	99.83±0.13
13	97.47±1.06	96.76±1.96	94.77±5.34	99.34±0.55	96.24±2.41	94.52±2.22	97.95±0.98	86.71±2.66	**99.51±0.65**	78.38±4.55	98.79±0	97.48±0.46
14	88.05±6.86	90.35±2.13	95.76±3.99	89.78±3.89	94.72±3.43	97.22±2.52	94.18±8.23	74.41±4.53	**98.14±1.53**	83.36±5.31	94.66±1.41	94.60±2.58
15	53.08±5.47	52.21±5.07	48.60±34.21	84.11±11.60	91.42±1.70	91.51±1.88	88.86±6.24	78.25±2.48	98.30±1.25	66.92±5.31	91.47±0.61	**98.70±0.37**
16	88.47±5.68	72.58±11.05	89.31±6.09	**99.90±0.20**	99.63±0.41	75.19±13.01	99.00±1.38	99.88±0.08	99.52±0.43	97.15±1.35	95.19±2.98	98.92±0.92
OA	87.64±0.61	83.48±0.81	85.75±5.01	95.11±1.86	96.70±0.43	88.77±2.21	93.20±0.82	90.87±1.25	98.87±0.64	86.18±1.36	96.40±0.43	**99.14±0.26**
AA	90.35±0.35	87.60±2.01	88.87±4.24	96.18±0.70	97.23±0.79	91.57±2.32	95.46±0.55	93.21±1.46	**99.19±0.30**	89.66±1.96	97.04±0.34	98.78±0.91
Kappa	86.18±0.69	81.57±0.93	84.09±5.56	94.57±2.06	96.34±0.47	87.54±2.44	92.43±0.92	89.89±1.38	98.74±0.72	84.65±1.51	95.99±0.47	**99.05±0.28**

TABLE VI

CLASSIFICATION ACCURACY (%) OF ALL ALGORITHMS ON THE HOUSTON2013 DATASET

Class	ML-based		CNN-based			Transformer-based			GNN-based			
	SVM	KNN	2DCNN	DBDA	HybridSN	ViT	SpectralFormer	MAT-ASSAL	CEGCN	MiniGCN	MDGCN	FRFENet
1	98.09±0.86	96.77±2.00	97.57±2.06	97.53±4.34	96.30±4.01	91.78±4.13	94.49±6.12	97.87±1.96	**98.93±1.10**	98.29±0.07	96.46±1.78	92.91±5.06
2	85.30±7.68	81.18±4.25	40.86±24.87	81.9±22.89	60.42±8.12	48.94±18.24	37.41±14.36	**95.77±3.18**	68.54±22.59	89.12±6.62	91.32±6.66	81.37±13.66
3	29.54±26.00	54.08±13.13	29.03±13.13	69.42±16.37	37.13±5.46	36.59±12.64	47.70±13.04	**88.58±4.85**	86.45±8.50	76.27±9.97	78.89±12.65	86.71±6.40
4	97.08±2.13	96.15±3.36	96.90±2.63	97.91±1.56	90.01±7.76	75.63±16.50	67.99±12.81	**99.28±3.03**	97.54±3.08	94.32±1.25	96.14±2.45	92.03±5.38
5	97.74±4.14	88.78±8.76	99.32±1.36	90.51±8.15	97.21±2.24	78.21±18.43	93.92±6.42	**99.69±1.97**	93.78±10.07	99.18±0.77	97.38±1.63	97.78±3.19
6	84.77±6.11	82.08±8.54	70.63±7.38	94.48±7.28	73.33±11.15	25.99±7.38	37.30±3.69	**96.28±3.44**	86.22±7.50	85.07±0.24	82.31±6.61	86.07±5.25
7	74.96±23.56	75.62±17.67	38.70±16.70	72.07±18.74	65.07±12.28	20.31±3.25	28.47±6.89	96.80±1.91	70.78±23.88	84.91±20.43	89.01±14.10	**99.91±0.26**
8	38.54±20.45	24.81±8.96	49.12±10.90	64.55±33.53	72.28±5.43	27.30±12.66	27.16±12.98	**92.93±1.78**	51.94±14.67	50.07±7.46	64.08±5.44	48.10±13.36
9	69.81±13.66	64.76±7.64	83.91±10.41	79.71±14.02	82.11±0.73	82.58±14.37	89.53±2.20	88.21±2.11	86.7±11.18	27.94±11.53	73.64±12.03	**89.67±9.43**
10	12.55±7.81	16.47±7.39	57.81±11.63	76.02±8.81	37.13±3.09	25.69±25.45	21.32±6.30	94.67±2.34	85.28±5.98	34.35±5.56	91.73±1.09	**94.78±3.98**
OA	71.61±4.13	70.07±2.59	73.25±5.05	82.85±8.31	75.76±0.87	58.09±4.70	61.20±0.58	**95.14±2.20**	85.12±3.44	73.86±2.38	86.50±0.17	86.93±2.38
AA	68.84±5.15	68.07±2.89	66.39±6.51	70.40±1.18	71.22±1.77	51.30±5.69	54.53±0.75	**95.01±2.65**	82.62±4.20	73.95±2.83	86.10±0.93	86.93±6.60
Kappa	67.96±4.71	66.22±2.98	69.63±5.82	70.81±1.85	72.59±1.02	52.27±5.58	55.98±0.65	**94.53±2.51**	83.22±3.89	70.71±2.67	84.83±0.17	85.33±2.65

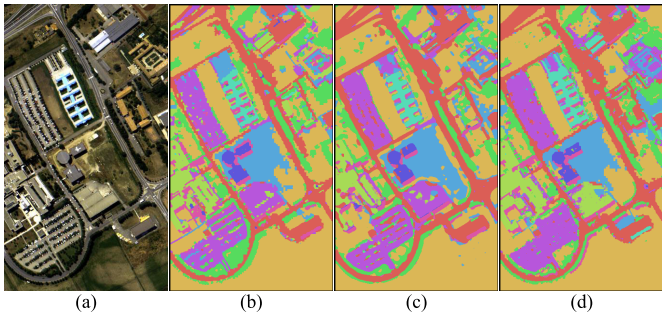
accuracy of 98.87%. However, CEGCN only extracts pixel-level and node-level features from local- and short-range scopes, while the proposed algorithm also extracts spectral information of ground objects from long-range scopes, resulting in the highest classification accuracy.

To further evaluate the classification performance of the FRFENet algorithm on complex hyperspectral images, the Houston2013 dataset was selected in this section, characterized by a complex distribution of ground objects, varying object sizes, and limited labeled samples, as shown in Fig. 11(a). Table VI shows the experimental results of all methods. Compared with the general method, the classification performance of FRFENet has improved significantly. However, compared with MAT-ASSAL, FRFENet's performance is lower. The reasons are as follows. The ground object distribution of the Houston2013 dataset is complicated, FRFENet does not give much thought to the scale of superpixel segmentation. For MAT-ASSAL, when the segmentation scale is appropriate, adaptive superpixel segmentation can better assist the promotion of pseudolabels. However, FRFENet is easier to implement, and the number of parameter runs is superior. At the same time, FRFENet avoids more hyperparameters. It is worth noting that most comparison methods have low classification accuracy for Class 10. This is because the Houston2013 dataset has a very small number of Class 10, and the feature distribution is extremely complex, so it can be confused with other classes. It is impossible to carry out effective feature extraction for this class only by relying on a small number of labeled samples. After the sample enhancement by balancing the quantity and quality, the increase of samples makes this class can be effectively identified. QQBSEsNet can extend the sample effectively, and the classification accuracy of Class 10 is improved effectively. This further explains the effectiveness and rationality of FRFENet.

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TABLE VII
OA (%) OF EACH BRANCH OF FRFESNET

Dataset	SLGConv	CNN	FRFEsNet (SLGConv+CNN)
PaviaU	98.44±0.43	97.74±0.55	**98.72±0.41**
Salinas	98.75±0.36	98.17±0.41	**99.14±0.26**
Houston2013	86.70±2.62	84.44±2.40	**86.93±2.38**



(a) (b) (c) (d)

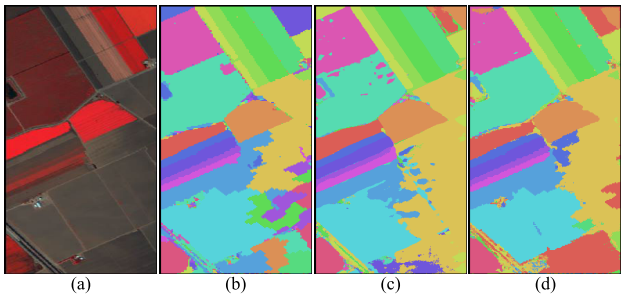
Fig. 12. Classification result maps of different branches of FRFEsNet on the PaviaU dataset. (a) False-color image. (b) SLGConv. (c) CNN. (d) FRFEsNet (SLGConv+CNN).

D. Discussion of Experimental Parameters

In this section, the effectiveness of the FRFENet is first verified. Then, the influence of different numbers of training samples on the classification performance of each algorithm is discussed. Moreover, the analysis is carried out on the impact of the QQBSEsNet on the classification performance. Finally, a comparison is made regarding the training time and NPs of each algorithm.

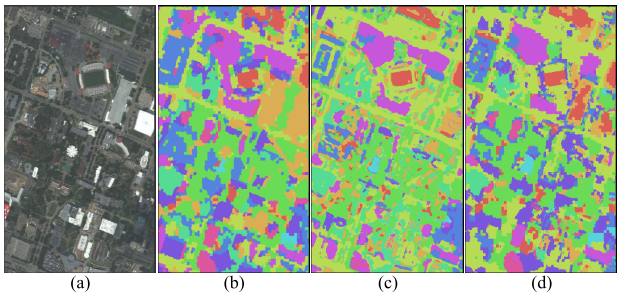
1) Influence of the FRFEsNet: The FRFEsNet combines local-range, short-range, and long-range spatial–spectral features to achieve full-range feature extraction. FRFEsNet has two branches, and therefore, in this section, we primarily evaluate the impact of these two branches on the classification performance. In addition, the NPs are set consistent with the parameters used in Sections III-A and III-B. Table VII displays the OA achieved by each branch of FRFEsNet. In the validation experiments, the complete framework of FRFENet was utilized, with the exception of substituting FRFEsNet.

As indicated in Table VII, the classification accuracy of FRFEsNet is significantly higher than that of individual branches. Specifically, for the Salinas dataset, FRFEsNet exhibits an improvement of 0.39% and 0.97% in OA compared to SLGConv and CNN, respectively. Table VII demonstrates that FRFEsNet is capable of integrating the abilities of SLGConv and CNN to extract features at different ranges. To provide a more detailed illustration of the benefits of FRFEsNet, the classification result maps of the FRFENet framework utilizing SLGConv, CNN, and FRFEsNet are presented in Figs. 12–14. The classification result maps above demonstrate that CNN can only extract local features of ground objects, which may lead to over-smoothing and less pronounced object edge structures. On the other hand, SLGConv can extract both long-range and short-range features, effectively addressing the spectral variability of hyperspectral data and maintaining consistent classification



(a) (b) (c) (d)

Fig. 13. Classification result maps of different branches of FRFEsNet on the Salinas dataset. (a) False-color image. (b) SLGConv. (c) CNN. (d) FRFEsNet (SLGConv+CNN).



(a) (b) (c) (d)

Fig. 14. Classification result maps of different branches of FRFEsNet on the Houston2013 dataset. (a) False-color image. (b) SLGConv. (c) CNN. (d) FRFEsNet (SLGConv+CNN).

TABLE VIII
OA (%) OF EACH BRANCH OF QQBSESNET

Dataset	None	Only Increasing Quantity	QQBSEsNet
PaviaU	93.50±1.27	96.79±1.03	**98.72±0.41**
Salinas	95.75±0.73	98.50±0.81	**99.14±0.26**
Houston2013	82.30±3.48	85.67±3.11	**86.93±2.38**

results for global ground objects. However, due to the inability of SLGConv to perceive object features within local distances, the classification results may lack smoothness and may not accurately reflect the actual distribution of ground objects. In contrast, Figs. 12–14(d) shows that the classification results of FRFEsNet (SLGConv+CNN) ensure the accuracy of ground object classification, demonstrating the superiority of FRFEsNet in handling spectral variability, especially when the training samples are limited.

2) Influence of the QQBSEsNet: As indicated in Table VIII, the results of ablation experiments on QQBSEsNet are presented. Here, we conducted experiments on no data enhancement, considering only quantity enhancement and the proposed QQBSEsNet, respectively. Obviously, the OA of using QQBSEsNet is improved by about 5% on average than without using QQBSEsNet on three datasets. When only quantitative enhancement is performed, the classification accuracy is higher than that without data enhancement, which is very reasonable and in line with common sense. In particular, in QQBSEsNet, when quantity and quality are considered at the same time, the classification accuracy is further increased. This verifies that QQBSEsNet does not simply increase duplicate samples, but increases a sufficient number

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TABLE IX
RUNNING TIME (S) AND NUMBER OF PARAMETERS OF ALL ALGORITHMS

Dataset	Times	CNN-based			Transformer-based			GNN-based			
		2DCNN	DBDA	HybridSN	ViT	SpectralFormer	MAT-ASSAL	CEGCN	MiniGCN	MDGCN	FRFENet
PaviaU	Training	43	74.67	40.57	859.51	883.11	207.2	42.86	95.89	7.51	80.84
	Testing	4.23	13.94	23.72	5.01	5.07	33.1	4.58	19.39	1.31	0.05
	NP	123.79K	202.77K	4017.40K	174.45K	175.21K	825.21K	152.86K	15.19K	6.81K	1125.02K
Salinas	Training	112	166.56	50.77	1090.73	1090.17	297.7	24.03	173.54	3.57	36.76
	Testing	7.17	33.73	29.76	6.01	6.35	47.3	2.75	42.48	0.10	0.02
	NP	153.78K	389.64K	4018.30K	174.90K	175.66K	884.75K	166.89K	29.22K	13.31K	638.64K
Houston2013	Training	24	27.85	22.2	94.72	102.72	247.2	40.74	5.08	3.25	70.58
	Testing	0.99	1.92	2.82	0.53	0.52	34.6	4.19	0.15	0.06	0.04
	NP	135.72K	279.48K	4017.53K	174.51K	175.27K	849.02K	158.32K	20.65K	9.33K	975.48K

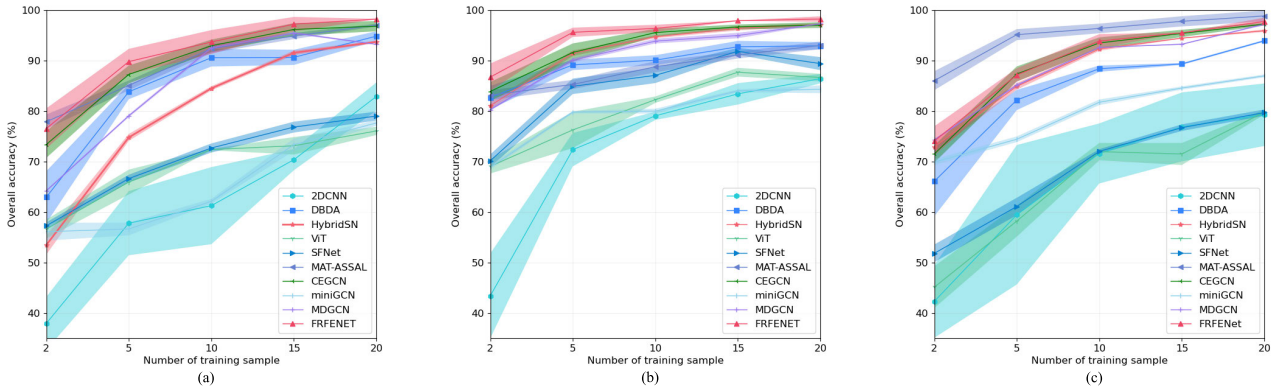


Fig. 15. Mean and standard deviation of the OA of all algorithms under different numbers of training samples, where the width of the shaded part represents the size of the standard deviation and the solid line represents the mean value of OA. (a) PaviaU. (b) Salinas. (c) Houston2013.

of high-quality samples. This fully shows that QQBSEsNet can effectively expand the sample to improve the classification accuracy.

3) Influence of the Training Sample Size: At present, the number of available training samples greatly affects the classification performance of algorithms. To assess the classification accuracy of FRFENet with a limited number of training samples, we conducted experiments using 2, 5, 10, 15, and 20 samples from each category as the training dataset. In addition, one sample from each category was set aside for validation purposes. The average OAs of the evaluated algorithms, including FRFENet, are presented in Fig. 15.

The results provide insights into the algorithm's performance under varying training sample sizes and shed light on its robustness and effectiveness. It is evident that FRFENet outperformed the other nine algorithms in terms of classification accuracy across all three datasets. Notably, the superior performance of FRFENet becomes more pronounced as the number of training samples decreases. Even with only two training samples per category, FRFENet achieved an OA of 76.54% on the PaviaU dataset. The results depicted in Fig. 15 demonstrate the potential of FRFENet in handling HSIC tasks with limited labeled samples. This success can be attributed to its innovative features, such as the FRFENet and the quality–quantity balance approach for pseudolabeled samples. These techniques contribute to improved consistency, accuracy, and robustness in the classification process.

4) Analysis of Runtime and Network Parameters: In this section, a comparison of performance parameters is conducted for the deep learning algorithm, including training time, testing time, and the number of NPs. These results are presented in Table IX. Table IX demonstrates that the training time of the FRFENet algorithm is not significantly increased compared with the CNN-based and GCN-based algorithms. It is roughly comparable with the training time of mainstream algorithms. At the same time, the training time of the proposed algorithm is an order of magnitude lower than that of the transformer-based algorithm. FRFENet algorithm achieves faster training and testing time while maintaining classification performance. The network model of the FRFENet algorithm includes GCN and CNN, resulting in more NPs compared with typical algorithms, but still fewer than the parameters of the HybridSN algorithm. However, the increased number of NPs does not slow down the algorithm's runtime.

IV. CONCLUSION

In this article, an FRFENet based on quality–quantity-balanced sample enhancement is proposed for hyperspectral imagery classification. This article presents a novel approach called FRFENet that leverages quality–quantity-balanced sample enhancement for improved classification performance. The proposed framework ensures consistency and accuracy of class features across the entire range by incorporating a full-range feature extraction method. This method combines local-range, short-range, and long-range spatial–spectral features to address spectral variability, especially in scenarios with limited labeled samples. Furthermore, the quality–quantity balance approach enhances the quantity and quality of pseudolabeled samples, effectively utilizing unlabeled data. In addition, the utilization of superpixel region homogeneity increases the number of

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training samples, thereby further enhancing the algorithm's classification performance. Experimental results conducted on three datasets demonstrate that FRFENet has advanced classification performance.

The proposed method can effectively overcome the challenges posed by spectral variability and limited labeled samples. At the same time, FRFENet avoids more hyperparameters. These findings highlight the potential of FRFENet in various high-dimensional remote sensing applications, offering improved accuracy and robustness in classification tasks. Compared with the existing mainstream methods, the proposed method cannot achieve the optimal running time and the number of NPs. Therefore, future research will focus on work related to lightweight models to speed up operation and reduce the number of parameters.

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